Practical No. 07

Aim : Create camera shake effect in unity

Step 1: Create the script file

1. Open your Unity project.
2. In the **Project window** (bottom of Unity), right-click in the **Assets** folder.
3. Select **Scripting → Empty C# Script**.
4. Name it **CameraShake**.

Step 2: Open the script

1. In the **Project window**, double-click on the CameraShake script you just created.
2. It will open in **Visual Studio** (or your default code editor).

Step 3: Make it a Unity script

C# Code: Copy paste this code and ctrl+s

```
using UnityEngine;

public class CameraShake : MonoBehaviour
{
    // Transform of the camera to shake
    public Transform camTransform;

    // How long the object should shake for
    public float shakeDuration = 0f;

    // Amplitude of the shake. A larger value shakes the camera harder
    public float shakeAmount = 0.7f;

    // How quickly the shaking stops
    public float decreaseFactor = 1.0f;

    private Vector3 originalPos;

    void Awake()
{
    if (camTransform == null)
        camTransform = transform;

    // Save the original position here
    originalPos = camTransform.localPosition;
}
```

```
    void Update()
{
    if (shakeDuration > 0)
    {
        camTransform.localPosition = originalPos + Random.insideUnitSphere * shakeAmount;
        shakeDuration -= Time.deltaTime * decreaseFactor;
    }
    else
    {
        shakeDuration = 0f;
        camTransform.localPosition = originalPos;
    }

    // Press Space to test shake
    if (Input.GetKeyDown(KeyCode.Space))
    {
        shakeDuration = 0.5f;   // how long the shake lasts
        shakeAmount = 0.3f;     // how strong the shake is
    }
}

}
```

Step 4: Attach the script to the camera

1. Go back to **Unity**.
2. In the **Hierarchy**, click on your **Main Camera**.
3. In the **Inspector** (on the right side), scroll down.
4. Click **Add Component** → search for **CameraShake** → Click it/Press Enter.

Step 5: Set up values to test the shake

In the **Inspector** (where you see **CameraShake (Script)** on your Main Camera):

- **Shake Duration** → set it to 1.
- **Shake Amount** → set it to 0.3.
- **Decrease Factor** → leave it at 1.

Step 6: Add Camera inside the Scene

Select your **Main Camera** in the Hierarchy.

In the Inspector under **CameraShake (Script)**, you'll see a field called **Cam Transform**.
Drag the **Main Camera** itself from the Hierarchy into that slot.

Step 7: Add a Cube or any shape as a visual reference

1. In the **Hierarchy**, right-click → **3D Object** → **Cube**.
2. You'll now see a cube in your scene.
3. Position it so the camera can see it:

Step 8: Create & Apply a Material in Unity

1. Go to the **Project window** (bottom area where Assets are).
2. **Right-click** → **Create** → **Material**.
3. Name it something like **"SphereMat"** or **"CubeMat"**.
4. In the Inspector (on the right), you'll see the Material settings.
   - Change the **Surface Input → Base Map** (pick any color) and press enter.
5. Now, drag your new **SphereMat** from the Project window onto your **Cube** in the Scene or Hierarchy.

Step 9: Make sure your Cube uses the material

1. Select your **Cube**.
2. In Inspector, find **Mesh Renderer** → **Materials** → **Element 0**.
3. Drag your colored material (**SphereMat**) onto **Element 0**.

Step 10: Now, click the **Play** button at the top of Unity.

OUTPUT:

Output Video: Click here to watch
[Click here to watch the output video]

Output Image: